

Yaw Obeng Frimpong – Project 1

1. Steps

For the steps cost, my heuristic counts how many letter positions in the current word are different from the goal.

$h(n)$ = number of mismatched positions

For example, dog and cat have 3 mismatched positions, so $h(n) = 3$. This is a lower bound because one action can only change one letter. Every wrong position has to be changed at least once, and each change costs 1. More moves might be needed to make legal words, but fewer than the number of mismatches are not possible. Therefore, the heuristic is admissible.

Scrabble

For the Scrabble cost, my heuristic adds the Scrabble values of the goal letters that still need to be placed in their correct positions.

$h(n)$ = sum of ScrabbleValue(goal[i]) for each mismatched position

For example, if a position needs to become c, the heuristic adds 3 because c has a Scrabble value of 3. These costs are unavoidable because every incorrect position eventually has to receive its correct goal letter. A solution might need extra letter changes to make legal words, which would only increase the total cost. Therefore, this heuristic does not overestimate and is admissible.

Frequency

For the frequency cost, my heuristic counts the number of mismatched positions.

$h(n)$ = number of mismatched positions

A frequency move costs:

$1 + \text{rarity}(\text{word})$

Since rarity is nonnegative, every move costs at least 1. Also, one move can only change one letter. So if there are 3 mismatched positions, at least 3 moves are needed and the remaining cost must be at least 3. The heuristic ignores the extra rarity costs, so the real cost can be higher, but the heuristic cannot overestimate it. Therefore, it is admissible.

2.

Word Pair	Cost Measure	UCS cost	UCS nodes expanded	A* cost	A* Nodes expanded	Reduction
WHY → ASK	Steps	5	773	5	16	97.93%
WHY →	Scrabble	9	765	9	13	98.3%

ASK						
WHY → ASK	Frequency	38.8554	808	38.8554	755	6.56%
BEAR → DUCK	Steps	4	691	4	6	99.13%
BEAR → DUCK	Scrabble	11	2981	11	6	99.80%
BEAR → DUCK	Frequency	38.2402	1084	38.2402	720	33.58%

1. Yes. They returned the same optimal cost in all six tests.
2. Largest reduction:
For WHY → ASK, Scrabble had the largest reduction at 98.30%. For BEAR → DUCK, Scrabble was also the largest at 99.80%.
3. Least informative:
Frequency was the least informative for both problems. It only reduced expansions by 6.56% and 33.58%.
4. The steps heuristic uses the number of mismatched letters, which gives A* a good estimate of how many changes are still needed. The Scrabble heuristic is more specific because it uses the values of the goal letters that still need to be inserted, which helped A* expand even fewer nodes. The frequency heuristic only uses mismatched letters and ignores the rarity part of the cost, so it gives A* less information and causes it to expand more nodes.

3. Steps — Yes

The cost is the same in both directions because every move costs 1. Reversing a path gives the same number of moves.

Scrabble — No

The cost depends on the new letter being inserted, so reversing a move can have a different cost.

Example:

WHY → ASK = 9

ASK → WHY = 14

Frequency — No

The cost depends on the rarity of the word being entered.

Example:

DOG $\rightarrow$ COG = 12.39

COG $\rightarrow$ DOG = 6.82

So, an action can be reversible without having the same cost in both directions.

4. **AI used:** DeepSeek**Prompt**

In DOGCAT, the initial and goal states are lowercase words of the same length—three or four letters—and every legal state must appear in the supplied dictionary. A legal action changes exactly one letter. Under the frequency objective, entering word w costs $1 + \text{rarity}(w)$, where the nonnegative rarity value is stored in the dictionary.

The baseline heuristic $h_0(n)$ is the number of positions in which state n differs from the goal. Propose a rarity-aware heuristic intended to be more informative than h_0 while remaining admissible. Define it precisely and justify your proposal. By “more informative,” we mean that the proposed heuristic should be at least as large as h_0 for every state and strictly larger for at least one state, without overestimating the true optimal remaining cost.

Proposed Heuristic

DeepSeek proposed adding rarity information to h_0 :

- Goal: $h(n) = 0$
- One mismatch: $h(n) = h_0(n) + \text{rarity}(\text{goal})$
- Two or more mismatches: $h(n) = h_0(n) + \text{rarity}(\text{goal}) + \text{minimum intermediate rarity}$

Audit

The heuristic is admissible because every solution must enter the goal, and with two or more mismatches, at least one intermediate word is also needed. DeepSeek made one mistake by saying the minimum rarity is positive if any word has positive rarity. In our dictionary, however, every rarity is positive, with a minimum of **0.016226212**, so the heuristic works for this problem.

Results

Problem	h_0 Expanded	New h Expanded	Optimal Cost
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WHY → ASK	755	634	38.8554
BEAR → DUCK	720	290	38.2402

The new heuristic expanded fewer nodes while keeping the same optimal costs, so it was more informative than h_0 .